THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Metals produce a ringing sound when struck. This property is called: 1

(a) lustre **(b)** malleability **(c)** sonority **(d)** ductility

2. The most ductile metal is: 1

(a) mercury **(b)** gold **(c)** lead **(d)** iron

3. Which metal melts if kept on the palm? 1

(a) sodium **(b)** zinc **(c)** lead **(d)** gallium

4. A student states: “Iodine is a non-metal and therefore cannot be lustrous.” The correct response is: 1

(a) Correct
(b) Incorrect — iodine is a lustrous non-metal
(c) Iodine is a metal
(d) Cannot be determined

5. The allotrope of carbon that is the hardest natural substance is: 1

(a) fullerene **(b)** graphite **(c)** coal **(d)** diamond

6. The number of electrons shared in a double bond is: 1

(a) 4 **(b)** 6 **(c)** 2 **(d)** 8

7. Carbon forms the largest number of compounds because of: 1

(a) high reactivity **(b)** metallic nature
(c) catenation and tetravalency **(d)** ionic bonding

8. The molecular formula of benzene is: 1

(a) C_6H_{14} **(b)** C_5H_{10} **(c)** C_6H_6 **(d)** C_6H_{12}

9. A student states: “Ethanol is a good conductor of electricity.” The correct response is: 1

(a) Correct **(b)** Correct when warm
(c) Incorrect — it does not ionise **(d)** Cannot be determined

10. Which of these will give a sooty flame? 1

(a) naphthalene **(b)** LPG **(c)** ethanol **(d)** methane

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why are metals called electropositive elements? 2

12. Give two examples each of metals that are 2
(i) malleable
(ii) ductile.

13. Why are carbon compounds generally poor conductors of electricity? 2

14. Name two allotropes of carbon and one use of each. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. What happens when 3
(i) zinc is heated in air
(ii) aluminium powder is heated with manganese dioxide
(iii) magnesium reacts with dilute HCl? Write equations.

16. State three differences between saturated and unsaturated hydrocarbons. 3

17. Describe an activity to show that metals are good conductors of electricity. 3

OR

Why do ionic compounds have high melting and boiling points? Give two examples with their melting points. 3

18. Write the names and formulae of the first three members of the alkyne series. 3

OR

How is the cleansing action of soap affected by hard water? Suggest a remedy. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is enrichment of ores? 5
(ii) Explain the steps of extracting zinc from zinc blende with equations.
(iii) Why is reduction done with carbon?

OR

Case study: Ravi placed iron nails in three test tubes: A with water exposed to air, B with boiled water covered by an oil layer, C with anhydrous calcium chloride (dry air). 5
(i) In which tube do nails rust?
(ii) Why is oil added in B?
(iii) What is the role of $CaCl_2$ in C?
(iv) What conclusion is drawn?
(v) Name one method to prevent rusting.

20. (i) Explain the nomenclature of carbon compounds with the help of prefixes/suffixes for halogen, alcohol, aldehyde and ketone. 5
(ii) Name: CH_3CH_2Br, CH_3CH_2CHO.

OR

Case study: A student mixed 1 mL ethanol and 1 mL glacial acetic acid with a few drops of conc. H_2SO_4 and warmed it in a water bath. A sweet smell was observed. 5
(i) Name the reaction.
(ii) Write the equation.
(iii) Role of conc. H_2SO_4?
(iv) What happens when the product is treated with NaOH?
(v) Give one use of the product.

END OF QUESTION PAPER Total: 40 Marks